

Lexib® Hard Capsule 200 mg

1 NAME OF THE MEDICINAL PRODUCT

Lexib® Hard Capsule 200 mg

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

Each capsule contains 200 mg celecoxib.

For full list of excipients, see section 6.1.

3 PHARMACEUTICAL FORM

A size #2 white color hard gelatin capsule with yellow stripe printed with "200" on cap and "YSP" on body, containing white to yellowish-white powder.

4 CLINICAL PARTICULARS

4.1 Therapeutic indications

- For the management of acute pain in adults and for the treatment of primary dysmenorrhoea.
- Relief of the acute and chronic pain and inflammation of rheumatoid arthritis and osteoarthritis.
- Relief of signs and symptoms of ankylosing spondylitis.
- For the management of low back pain (100 mg and 200 mg only).

4.2 Posology and method of administration

Celecoxib capsules can be taken with or without food.

Given the association between cardiovascular (CV) risk and exposure to COX-2 Inhibitors, doctors are advised to use the lowest effective dose for the shortest possible duration of treatment.

Adults

Symptomatic Treatment of Osteoarthritis (OA): The recommended dose of celecoxib is 200 mg administered as a single dose or as 100 mg twice per day.

Symptomatic Treatment of Rheumatoid Arthritis (RA): The recommended dose of celecoxib is 100 mg or 200 mg twice per day.

Ankylosing Spondylitis (AS): The recommended dose of celecoxib is 200 mg administered as a single dose or 100 mg twice per day. Some patients may benefit from a total daily dose of 400 mg.

Management of Acute Pain: The recommended dose of celecoxib is 400 mg, initially, followed by an additional 200 mg dose, if needed on the first day. On subsequent days, the recommended dose is 200 mg twice daily, as needed.

Treatment of Primary Dysmenorrhea: The recommended dose of celecoxib is 400 mg, initially, followed by an additional 200 mg dose, if needed on the first day. On subsequent days, the recommended dose is 200 mg twice daily, as needed.

Low Back Pain (LBP): Usual dosage for adults is 100 mg of celecoxib orally twice daily, morning and evening, or 200 mg once daily (100 mg and 200 mg only).

* Lexib® Hard Capsule 200 mg is available at the strength of 200 mg only and may not be able to deliver all the dosing recommendations mentioned above. In such cases, other approved strengths should be used.

Special Populations

Elderly

No dosage adjustment is generally necessary. However, for elderly patients with a lower than average body weight (<50 kg), it is advisable to initiate therapy at the lowest recommended dose.

Hepatic Impairment

No dosage adjustment is necessary in patients with mild hepatic impairment (Child-Pugh Class A). Introduce celecoxib at half the recommended dose in arthritis or pain patients with moderate hepatic impairment (Child-Pugh Class B). Patients with severe hepatic impairment (Child-Pugh Class C) have not been studied.

Renal Impairment

No dosage adjustment is necessary in patients with mild or moderate renal impairment. There is no clinical experience in patients with severe renal impairment.

Paediatric population

Celecoxib is not indicated for use in children.

CYP2C9 Poor Metabolizers

Patients who are known, or suspected to be CYP2C9 poor metabolizers based on genotyping or previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution as the risk of dose-dependent adverse effects is increased. Consider reducing the dose to half the lowest recommended dose.

4.3 Contraindications

Celecoxib is contraindicated in:

- Patients with known hypersensitivity to celecoxib or any other ingredient of the product listed.
- Patients with known sulfonamide hypersensitivity.
- Patients who have experienced asthma, urticaria or allergic-type reactions after taking acetylsalicylic acid (ASA [aspirin]) or other nonsteroidal anti-inflammatory drugs (NSAIDs), including other cyclooxygenase-2 (COX-2) specific inhibitors.
- Treatment of peri-operative pain in the setting of coronary artery bypass graft (CABG) surgery.
- Patients who have established CV disease (ischaemic heart disease and stroke).

4.4 Special warnings and precautions for use

WARNING

RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Patients with risk factors of heart disease, hypertension, hyperlipidaemia, diabetes mellitus, smoking patients and patients with peripheral arterial disease should only be treated with celecoxib after careful consideration.

Cardiovascular Effects

Cardiovascular Thrombotic Events: Celecoxib may cause an increased risk of serious CV thrombotic events, myocardial infarction (MI), and stroke, which can be fatal. All NSAIDs may have a similar risk. This risk may increase with dose and duration of use. The relative increase of this risk appears to be similar in those with or without known CV disease or CV risk factors. However, patients with CV disease or CV risk factors may be at greater risk in terms of absolute incidence, due to their increased rate at baseline. To minimize the potential risk for an adverse CV event in patients treated with celecoxib, the lowest effective dose should be used for the shortest duration possible. Physicians and patients should remain alert for the development of such events, even in the absence of previous CV symptoms. Patients should be informed about the signs and symptoms of serious CV toxicity and the steps to take if they occur.

Two large, controlled, clinical trials of a different COX-2 selective NSAID for the treatment of pain in the first 10 to 14 days following CABG surgery found an increased incidence of myocardial infarction and stroke.

Celecoxib is not a substitute for acetylsalicylic acid for prophylaxis of CV thromboembolic diseases because of the lack of effect on platelet function. Because celecoxib does not inhibit platelet aggregation, anti-platelet therapies (e.g., acetylsalicylic acid) should not be discontinued.

Hypertension: As with all NSAIDs, celecoxib can lead to the onset of new hypertension or worsening of pre-existing hypertension, either of which may contribute to the increased incidence of CV events. Patients taking angiotensin converting enzyme (ACE) inhibitors, thiazide diuretics or loop diuretics may have impaired response to these therapies when taking NSAIDs. NSAIDs, including celecoxib, should be used with caution in patients with hypertension. Blood pressure should be monitored closely during the initiation of therapy with celecoxib and throughout the course of therapy.

Fluid Retention and Edema: As with other drugs known to inhibit prostaglandin synthesis, fluid retention and edema have been observed in some patients taking celecoxib. Therefore, patients with pre-existing congestive heart failure (CHF) or hypertension should be closely monitored. Celecoxib should be used with caution in patients with compromised cardiac function, pre-existing edema, or other conditions predisposing to, or worsened by, fluid retention including those taking diuretic treatment or otherwise at risk of hypovolemia.

Gastrointestinal (GI) Effects

Upper and lower GI perforations, ulcers or bleeds have occurred in patients treated with celecoxib. These serious adverse events can occur at any time, with or without warning symptoms, in patients treated with Celecoxib. Patients most at risk of developing these types of GI complications with NSAIDs are the elderly, patients with CV disease, patients using concomitant glucocorticoids, antiplatelet drugs (such as aspirin), or other NSAIDs, patients using alcohol or patients with a prior history of, or active, GI disease, such as ulceration, GI bleeding or inflammatory conditions. Other factors that increase the risk of GI bleeding in patients treated with NSAIDs include longer duration of NSAID therapy; concomitant use of oral corticosteroids, antiplatelet drugs (such as aspirin), anticoagulants; or selective serotonin reuptake inhibitors (SSRIs); smoking; use of alcohol; older age; and poor general health status. Most

spontaneous reports of fatal GI events have been in elderly or debilitated patients.

Strategies to Minimize the GI Risks in NSAID-treated Patients:

- Use the lowest effective dosage for the shortest possible duration.
- Avoid administration of more than one NSAID at a time.
- Avoid use in patients at higher risk unless benefits are expected to outweigh the increased risk of bleeding. For such patients, as well as those with active GI bleeding, consider alternate therapies other than NSAIDs.
- Remain alert for signs and symptoms of GI ulceration and bleeding during NSAID therapy.
- If a serious GI adverse event is suspected, promptly initiate evaluation and treatment, and discontinue Celecoxib until a serious GI adverse event is ruled out.
- In the setting of concomitant use of low-dose aspirin for cardiac prophylaxis, monitor patients more closely for evidence of GI bleeding.

Renal Effects

NSAIDs including celecoxib may cause renal toxicity. Clinical trials with celecoxib have shown renal effects similar to those observed with comparator NSAIDs. Patients at greatest risk for renal toxicity are those with impaired renal function, heart failure, liver dysfunction, and the elderly. Such patients should be carefully monitored while receiving treatment with celecoxib.

Caution should be used when initiating treatment in patients with dehydration. It is advisable to rehydrate patients first and then start therapy with celecoxib.

Advanced Renal Disease

Renal function should be closely monitored in patients with advanced renal disease who are administered celecoxib.

Anaphylactoid Reactions

As with NSAIDs in general, anaphylactoid reactions have occurred in patients exposed to celecoxib.

Serious Skin Reactions

Serious skin reactions, some of them fatal, including drug reaction with eosinophilia and systemic symptoms (DRESS syndrome), exfoliative dermatitis, Stevens-Johnson syndrome, and toxic epidermal necrolysis, have been reported very rarely in association with the use of celecoxib. Patients appear to be at highest risk for these events early in the course of therapy, the onset of the event occurring in the majority of cases within the first month of treatment. Celecoxib should be discontinued at the first appearance of skin rash, mucosal lesions, or any other sign of hypersensitivity.

Hepatic Effects

Patients with severe hepatic impairment (Child-Pugh Class C) have not been studied. The use of celecoxib in patients with severe hepatic impairment is not recommended. Celecoxib should be used with caution when treating patients with moderate hepatic impairment (Child-Pugh Class B), and initiated at half the recommended dose.

Rare cases of severe hepatic reactions, including fulminant hepatitis (some with fatal outcome), liver necrosis, and hepatic failure (some with fatal outcome or requiring liver transplant), have been reported with celecoxib.

A patient with symptoms and/or signs of liver dysfunction, or in whom an abnormal liver function test has occurred, should be monitored carefully for evidence of the development of a more severe hepatic reaction while on therapy with celecoxib.

Use with Oral Anticoagulants

The concomitant use of NSAIDs with oral anticoagulants increases the risk of bleeding and should be given with caution. Oral anticoagulants include warfarin/coumarin-type and novel oral anticoagulants (e.g. apixaban, dabigatran, and rivaroxaban). In patients on concurrent therapy with warfarin or similar agents, serious bleeding events, some of them fatal, have been reported. Because increases in prothrombin time (INR) have been reported, anticoagulation/INR should be monitored in patients taking a warfarin/coumarin-type anticoagulant after initiating treatment with celecoxib or changing the dose.

General

By reducing inflammation, celecoxib may diminish the utility of diagnostic signs, such as fever, in detecting infections.

The concomitant use of celecoxib and a non-aspirin NSAID should be avoided.

CYP2D6 Inhibition

Celecoxib inhibits CYP2D6. Although it is not a strong inhibitor of this enzyme, a dose reduction may be necessary for individually dose-titrated medicinal products that are metabolised by CYP2D6.

4.5 Interaction with other medicinal products and other forms of interaction

General: Celecoxib metabolism is predominantly mediated via cytochrome P450 (CYP) 2C9 in the liver. Patients who are known or suspected to be poor CYP2C9 metabolizers based on previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution as they may have abnormally high

plasma levels due to reduced metabolic clearance. Consider starting treatment at half the lowest recommended dose.

Concomitant administration of celecoxib with inhibitors of CYP2C9 can lead to increases in plasma concentrations of celecoxib. Therefore, a dose reduction of celecoxib may be necessary when celecoxib is co-administered with CYP2C9 inhibitors.

Concomitant administration of celecoxib with inducers of CYP2C9, such as rifampicin, carbamazepine and barbiturates can lead to decreases in plasma concentrations of celecoxib. Therefore, a dose increase of celecoxib may be necessary when celecoxib is co-administered with CYP2C9 inducers.

Clinical pharmacokinetics study and *in vitro* studies indicate that celecoxib, although not a substrate, is an inhibitor of CYP2D6. Therefore, there is a potential for an *in vivo* drug interaction with drugs that are metabolized by CYP2D6.

Drug-specific

Interaction of celecoxib with warfarin or similar agents: (See Section 4.4 Special warnings and precautions for use – Use with Oral Anticoagulants).

Lithium: In healthy subjects, lithium plasma levels increased approximately 17% in subjects receiving lithium together with celecoxib. Patients on lithium treatment should be closely monitored when celecoxib is introduced or withdrawn.

Aspirin: Celecoxib does not interfere with the anti-platelet effect of low-dose. Because of its lack of platelet effects, celecoxib is not a substitute for aspirin in the prophylactic treatment of CV disease.

Anti-hypertensives including Angiotensin-converting enzyme inhibitors (ACEIs), Angiotensin II antagonists (also known as angiotensin receptor blockers, [ARBs]), diuretics and beta-blockers: Inhibition of prostaglandins may diminish the effect of anti-hypertensives including ACEIs and/or ARBs, diuretics and beta-blockers. This interaction should be given consideration in patients taking celecoxib concomitantly with ACEIs and/or ARBs, diuretics and beta-blockers.

In patients who are elderly, volume-depleted (including those on diuretic therapy), or with compromised renal function, coadministration of NSAIDs, including selective COX-2 inhibitors, with ACE inhibitors, angiotensin II antagonists or diuretics, may result in deterioration of renal function, including possible acute renal failure. These effects are usually reversible. Therefore, the concomitant administration of these drugs should be done with caution. Patients should be adequately hydrated and the clinical need to monitor the renal function should be assessed at the beginning of the concomitant treatment and periodically thereafter.

Cyclosporine: Because of their effect on renal prostaglandins, NSAIDs may increase the risk of nephrotoxicity with cyclosporine.

Fluconazole and ketoconazole: Since celecoxib is predominantly metabolised by CYP2C9 it should be used at half the recommended dose in patients receiving fluconazole. Concomitant use of 200 mg single dose of celecoxib and 200 mg once daily of fluconazole, a potent CYP2C9 inhibitor, resulted in a mean increase in celecoxib C_{max} of 60% and in AUC of 130%. Ketoconazole, a CYP3A4 inhibitor, showed no clinically relevant inhibition in the metabolism of celecoxib.

Dextromethorphan and metoprolol: Concomitant administration of celecoxib 200 mg twice daily resulted in a 2.6-fold and a 1.5-fold increases in plasma concentrations of dextromethorphan and metoprolol (CYP2D6 substrates), respectively. These increases are due to celecoxib inhibition to the CYP2D6 substrate metabolism via CYP2D6. Therefore, the dose of drugs as CYP2D6 substrate may need to be reduced when treatment with celecoxib is initiated or increased when treatment with celecoxib is terminated.

Diuretics: NSAIDs, in some patients, can reduce the natriuretic effect of furosemide and thiazides by inhibition of renal prostaglandin synthesis.

Methotrexate: No pharmacokinetic and clinically important interactions have been observed in a clinical study between celecoxib and methotrexate.

Oral contraceptives: No clinically relevant effects on the pharmacokinetics of a prototype combination oral contraceptive (1 mg norethindrone/0.035 mg ethinyl estradiol).

Other drugs: No clinically important interactions between celecoxib and antacids (aluminium and magnesium), omeprazole, glibenclamide (glyburide), phenytoin, or tolbutamide.

4.6 Fertility, pregnancy and lactation

Fertility:

Based on the mechanism of action, the use of NSAIDs, including celecoxib, may delay or prevent rupture of ovarian follicles, which has been associated with reversible infertility in some women. In women who have difficulties conceiving or who are

undergoing investigation of infertility, withdrawal of NSAIDs, including celecoxib, should be considered.

Pregnancy:

There are no studies in pregnant women. Studies in animals have shown reproductive toxicity. The relevance of these data for humans is unknown.

Celecoxib, as with other drugs inhibiting prostaglandin synthesis, may cause uterine inertia and premature closure of the ductus arteriosus and should be avoided during the third trimester of pregnancy.

Celecoxib should be used during pregnancy only if the potential benefit to the mother justifies the potential risk to the fetus.

Inhibition of prostaglandin synthesis might adversely affect pregnancy. Data from epidemiological studies suggest an increased risk of spontaneous abortion after use of prostaglandin synthesis inhibitors in early pregnancy. In animals, administration of prostaglandin synthesis inhibitors has been shown to result in increased pre- and post-implantation loss.

If used during second or third trimester of pregnancy, NSAIDs may cause fetal renal dysfunction which may result in reduction of amniotic fluid volume or oligohydramnios in severe cases. Such effects may occur shortly after treatment initiation and are usually reversible upon discontinuation. Pregnant women on celecoxib should be closely monitored for amniotic fluid volume.

Lactation:

Studies in rats show that celecoxib is excreted in milk at concentrations similar to those in plasma. Administration of celecoxib to lactating women has shown very low transfer of celecoxib into breast milk. Because of the potential for adverse reactions in nursing infants from celecoxib, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the expected benefit of the drug to the mother.

4.7 Effects on ability to drive and use machines

The effect of celecoxib on ability to drive or use machinery has not been studied, but based on its pharmacodynamic properties and overall safety profile it is unlikely to have an effect.

4.8 Undesirable effects

Clinical Trials Experience

The following adverse drug reactions (ADRs) in Table 1 were identified with incidence rates greater than 0.01% in celecoxib group and greater than those reported in placebo group, during 12 placebo- and/or active-controlled clinical trials of treatment duration up to 12 weeks at daily doses from 100 mg up to 800 mg in adults.

The frequencies on the ADRs in Table 1 are updated based on a more recent pooling of 89 randomized, controlled clinical trials data representing clinical exposure in 38,102 patients taking celecoxib. ADR frequencies are defined as: very common ($\geq 10\%$), common ($\geq 1\%$ and $<10\%$), uncommon ($\geq 0.1\%$ and $<1\%$), rare ($\geq 0.01\%$ and $<0.1\%$) very rare ($<0.01\%$). The ADRs in Table 1 are listed by system organ class and ranked by frequency in descending order.

Table 1. Adverse Drug Reactions (ADRs) in 12 placebo- and/or active-controlled clinical trials and ADR frequency from 89 Pain and Inflammation Clinical Trials with Daily Doses of 25 mg-800 mg, in Adult Populations

System Organ Class Frequency	Adverse Drug Reaction
Infections and infestations Common Uncommon	Bronchitis, sinusitis, upper respiratory tract infection, urinary tract infection Pharyngitis, rhinitis
Blood and lymphatic system disorders Uncommon Rare	Anaemia Thrombocytopenia
Immune system disorders Uncommon	Hypersensitivity
Psychiatric disorders Common Uncommon Rare	Insomnia Anxiety Confusional state
Nervous system disorders Common Uncommon	Dizziness Hypertonia, somnolence
Eye disorders Uncommon	Vision blurred
Ear and labyrinth disorders Uncommon	Tinnitus
Cardiac disorders Uncommon Rare	Palpitations Cardiac failure congestive, arrhythmia, tachycardia
Vascular disorders Common Rare	Hypertension (including aggravated hypertension) Flushing

System Organ Class Frequency	Adverse Drug Reaction
Respiratory, thoracic and mediastinal disorders Common	Cough
Gastrointestinal disorders Common Uncommon Rare Very rare	Vomiting, abdominal pain, diarrhoea, dyspepsia, flatulence Gastric ulcer, tooth disorder Duodenal ulcer, oesophageal ulcer Intestinal perforation, pancreatitis
Hepatobiliary disorders Uncommon	Hepatic enzyme increased (includes alanine aminotransferase increased and aspartate aminotransferase increased)
Skin and subcutaneous tissue disorders Common Uncommon Rare Very rare	Pruritus (includes pruritus generalized), rash Urticaria, ecchymosis Angioedema, alopecia Dermatitis bullous
General disorders and administration site conditions Common Uncommon	Oedema peripheral edema Face oedema, influenza-like illness
Injury, poisoning and procedural conditions Uncommon	Injury

The following additional adverse drug reactions in Table 2 were identified with incidence rates greater than placebo in long-term polyp prevention studies of duration up to 3 years at daily doses from 400 mg up to 800 mg.

Frequencies of ADRs in Table 2 were determined based on these long-term polyp prevention studies and defined as: very common ($\geq 10\%$), common ($\geq 1\%$ and $< 10\%$), uncommon ($\geq 0.1\%$ and $< 1\%$). The ADRs in Table 2 are listed by system organ class and ranked by frequency in descending order.

Table 2. Adverse Reactions from Polyp Prevention Studies of Duration up to 3 Years and Daily Doses of 400 mg–800 mg

System Organ Class Frequency	Adverse Drug Reaction
Infections and infestations Common Uncommon	Ear infection, fungal infection** <i>Helicobacter</i> infection, herpes zoster, erysipelas, wound infection, gingivitis, labyrinthitis, bacterial infection
Neoplasms benign, malignant, and unspecified Uncommon	Lipoma
Psychiatric disorders Uncommon	Sleep disorder
Nervous system disorders Uncommon	Cerebral infarction
Eye disorders Uncommon	Conjunctival hemorrhage, vitreous floaters
Ear and labyrinth disorders Uncommon	Hypoacusis
Cardiac disorders Common Uncommon	Myocardial infarction, angina pectoris Angina unstable, aortic valve incompetence, arteriosclerosis coronary artery, sinus bradycardia, ventricular hypertrophy
Vascular disorders Very Common Uncommon	Hypertension* Deep vein thrombosis, haematoma
Respiratory, thoracic, and mediastinal disorders Common Uncommon	Dyspnoea Dysphonia
Gastrointestinal disorders Very Common Common Uncommon	Diarrhoea* Vomiting*, dysphagia, irritable bowel syndrome, gastrooesophageal reflux disease, nausea, diverticulum Hemorrhoidal haemorrhage, frequent bowel movements, mouth ulceration, stomatitis
Hepatobiliary disorders Common	Hepatic enzyme increased (includes alanine aminotransferase increased and aspartate aminotransferase increased)*
Skin and subcutaneous tissue disorders Uncommon	Dermatitis allergic
Musculoskeletal and connective tissue disorders Common Uncommon	Muscle spasms Synovial cyst

System Organ Class Frequency	Adverse Drug Reaction
Renal and urinary disorders Common Uncommon	Nephrolithiasis Nocturia
Reproductive system and breast disorders Common Uncommon	Vaginal haemorrhage, prostatitis, benign prostatic hyperplasia Ovarian cyst, menopausal symptoms, breast tenderness, dysmenorrhoea
General disorders and administration site conditions Uncommon	Oedema
Investigations Common Uncommon	Blood creatinine increased, prostatic specific antigen increased, weight increased Blood potassium increased, blood sodium increased, blood testosterone decreased, haematocrit decreased, haemoglobin increased
Injury, poisoning and procedural complications Uncommon	Foot fracture, lower limb fracture, fracture, epicondylitis, tendon rupture

*Hypertension, vomiting, diarrhoea, and hepatic enzyme increased are included in Table 2 because these events were reported more frequently in these studies, which were of 3-year duration, compared to Table 1, which includes adverse reactions from studies of 12-week duration.

**Fungal infections were primarily non-systemic.

Post-marketing Experience

Adverse reactions identified from post-marketing experience are provided below. Even though these were identified as reactions from post-marketing reports, trial data was consulted to estimate frequency. As above, frequencies are based on a pooling of trials representing exposure in 38,102 patients. Frequencies are defined as: very common ($\geq 10\%$), common ($\geq 1\%$ and $< 10\%$), uncommon ($\geq 0.1\%$ and $< 1\%$), rare ($\geq 0.01\%$ and $< 0.1\%$), very rare ($< 0.01\%$), not known (cannot be estimated from the available data):

Immune system disorders: Very rare: anaphylactic reaction

Psychiatric disorders: Rare: hallucination

Nervous system disorders: Very rare: cerebral haemorrhage, meningitis aseptic, ageusia, anosmia

Eye disorders: Uncommon: conjunctivitis

Vascular disorders: Very rare: vasculitis

Respiratory, thoracic and mediastinal disorders: Rare: pulmonary embolism, pneumonitis

Gastrointestinal disorders: Rare: gastrointestinal haemorrhage

Hepatobiliary disorders: Rare: hepatitis; Very rare: hepatic failure, hepatitis fulminant, hepatic necrosis, cholestasis, hepatitis cholestatic, jaundice

Skin and subcutaneous tissue disorders: Rare: photosensitivity reaction; Very rare: Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalised exanthematous pustulosis (AGEP), dermatitis exfoliative

Renal and urinary disorders: Rare: renal failure acute, hyponatremia; Very rare: tubulointerstitial nephritis, nephrotic syndrome, glomerulonephritis minimal lesion

Reproductive system and breast disorders: Rare: menstrual disorder; Not known: infertility female (female fertility decreased)

General disorders and administration site conditions: Uncommon: chest pain

4.9 Overdose

Clinical experience of overdose is limited. Single doses up to 1200 mg and multiple doses up to 1200 mg twice daily have been administered to healthy subjects without clinically significant adverse effects. In the event of suspected overdose, appropriate supportive medical care should be provided. Dialysis is unlikely to be an efficient method of drug removal because of high protein binding of the drug.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Non-steroidal anti-inflammatory and antirheumatic drugs, NSAIDs, Coxibs, ATC code: M01AH01

The mechanism of action of celecoxib is via inhibition of prostaglandin synthesis primarily by inhibition of cyclooxygenase 2 (COX-2). At therapeutic concentrations in humans, celecoxib does not inhibit cyclooxygenase 1 (COX-1). COX-2 is induced in response to inflammatory stimuli. This leads to the synthesis and accumulation of inflammatory prostanoids, in particular prostaglandin E₂, causing inflammation, edema and pain. Celecoxib acts as an anti-inflammatory, analgesic, and antipyretic agent in animal models by blocking the production of inflammatory prostanoids via COX-2 inhibition. In animal colon tumor models, celecoxib reduced the incidence and multiplicity of tumors.

In vivo and *ex vivo* studies show that celecoxib has a very low affinity for the constitutively expressed COX-1 enzyme. Consequently at therapeutic doses celecoxib has no effect on prostanoids synthesized by activation of COX-1 thereby not interfering

with normal COX-1 related physiological processes in tissues, particularly the stomach, intestine and platelets.

5.2 Pharmacokinetic properties

Absorption

When given under fasting conditions celecoxib is well absorbed reaching peak plasma concentrations after approximately 2-3 hours. Oral bioavailability from capsules is about 99% relative to administration in suspension (optimally available oral dosage form). Under fasting conditions, both peak plasma levels (C_{max}) and area under the curve (AUC) are roughly dose proportional up to 200 mg twice daily; at higher doses there are less than proportional increases in C_{max} and AUC.

Distribution

Plasma protein binding, which is concentration independent, is about 97% at therapeutic plasma concentrations and celecoxib is not preferentially bound to erythrocytes in the blood.

Metabolism

Celecoxib metabolism is primarily mediated via cytochrome P450 2C9. Three metabolites, inactive as COX-1 or COX-2 inhibitors, have been identified in human plasma: a primary alcohol, the corresponding carboxylic acid and its glucuronide conjugate.

Cytochrome P450 2C9 activity is reduced in individuals with genetic polymorphisms that lead to reduced enzyme activity, such as those homozygous for the CYP2C9*3 polymorphism.

In a pharmacokinetic study of celecoxib 200 mg administered once daily in healthy volunteers, genotyped as either CYP2C9*1/*1, CYP2C9*1/*3, or CYP2C9*3/*3, the median C_{max} and AUC₀₋₂₄ of celecoxib on day 7 were approximately 4-fold and 7-fold, respectively, in subjects genotyped as CYP2C9*3/*3 compared to other genotypes. In three separate single dose studies, involving a total of 5 subjects genotyped as CYP2C9*3/*3, single-dose AUC₀₋₂₄ increased by approximately 3-fold compared to normal metabolizers. It is estimated that the frequency of the homozygous *3/*3 genotype is 0.3 – 1.0% among different ethnic groups.

Patients who are known, or suspected to be CYP2C9 poor metabolizers based on previous history/experience with other CYP2C9 substrates should be administered celecoxib with caution. Consider starting treatment at half the lowest recommended dose.

Elimination

Elimination of celecoxib is mostly by hepatic metabolism with less than 1% of the dose excreted unchanged in urine. After multiple dosing, elimination half-life is 8 to 12 hours and the rate of clearance is about 500 mL/min. With multiple dosing steady-state plasma concentrations are reached before day 5. The intersubject variability on the main pharmacokinetic parameters (AUC, C_{max} , elimination half-life) is about 30%. The mean steady-state volume of distribution is about 500 L/70 kg in young healthy adults indicating wide distribution of celecoxib into the tissues. Preclinical studies indicate that the drug crosses the blood-brain barrier.

Food Effects

Dosing with food (high fat meal) delays absorption of celecoxib resulting in a T_{max} of about 4 hours and increases bioavailability by about 20%.

In healthy adult volunteers, the overall systemic exposure (AUC) of celecoxib was equivalent when celecoxib was administered as intact capsule or capsule contents sprinkled on applesauce. There were no significant alterations in C_{max} , T_{max} or $T_{1/2}$ after administration of capsule contents on applesauce.

Special Populations

Elderly

In the population >65 years there is a one and a half to two-fold increase in mean C_{max} and AUC for celecoxib. This is a predominantly weight-related rather than age-related change, celecoxib levels being higher in lower weight individuals and consequently higher in the elderly population who are generally of lower mean weight than the younger population. Therefore, elderly females tend to have higher drug plasma concentrations than elderly males. No dosage adjustment is generally necessary. However, for elderly patients with a lower than average body weight (<50 kg), initiate therapy at the lowest recommended dose.

Race

A meta-analysis of pharmacokinetic studies has suggested an approximately 40% higher AUC of celecoxib in the Black population compared to Caucasians. The cause and clinical significance of this finding is unknown.

Hepatic impairment

Plasma concentrations of celecoxib in patients with mild hepatic impairment (Child-Pugh Class A) are not significantly different from those of age and sex matched controls. In patients with moderate hepatic impairment (Child-Pugh Class B) celecoxib plasma concentrations are about twice those of matched controls.

Renal impairment

In elderly volunteers with age-related reductions in glomerular filtration rate (GFR) (mean GFR >65 mL/min/1.73 m²) and in patients with chronic stable renal insufficiency (GFR 35-60 mL/min/1.73 m²) celecoxib pharmacokinetics was comparable to those seen in patients with normal renal function. No significant relationship was found between serum creatinine (or creatinine clearance) and celecoxib clearance. Severe renal insufficiency would not be expected to alter clearance of celecoxib since the main route of elimination is via hepatic metabolism to inactive metabolites.

Renal effects

The relative roles of COX-1 and COX-2 in renal physiology are not completely understood. Celecoxib reduces the urinary excretion of PGE₂ and 6-keto-PGF_{1α} (a prostacyclin metabolite) but leaves serum thromboxane B₂ (TXB₂) and urinary excretion of 11-dehydro-TXB₂, a thromboxane metabolite (both COX-1 products) unaffected. Specific studies have shown Celecoxib produces no decreases in GFR in the elderly or those with chronic renal insufficiency. These studies have also shown transient reductions in fractional excretion of sodium. In studies in patients with arthritis a comparable incidence of peripheral edema has been observed to that seen with non-specific COX-inhibitors (which also possess COX-2 inhibitory activity). This was most evident in patients receiving concomitant diuretic therapy. However, increased incidences of hypertension and cardiac failure have not been observed and the peripheral edema has been mild and self-limiting.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Magnesium stearate, lactose, sodium lauryl sulfate, polyninylpyrrolidone, croscarmellose sodium, gelatin, titanium dioxide.

6.2 Incompatibilities

No incompatibilities are known.

6.3 Shelf life

3 years.

6.4 Special precautions for storage

Store below 30°C. Protect from light and moisture.

6.5 Nature and contents of container

Blister packing of 10's x 3.

Blister packing of 10's x 10.

6.6 Special precautions for disposal

Not applicable.

7 MANUFACTURER AND PRODUCT REGISTRATION HOLDER

Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)

Lot 3, 5, 7, 12 & 14, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43000 Kajang, Selangor, Malaysia.

Ordering Line: 1 800 88 3027

Product Info: 1 800 88 3679

8 DATE OF REVISION OF THE TEXT

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